

RUTU MAHESH GHATGE

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Summary

AI Engineer and AI & Data Science undergraduate with 2+ years of hands-on experience in Python backend development, computer vision, and NLP, building real-world AI systems using LLMs, CNNs, LSTMs, Kalman Filters, and YOLO. Internship experience across DRDO, Infosys Springboard, and AI startups, with strong skills in Python, TensorFlow, LangChain, and RAG architectures, along with exposure to IoT-based intelligent systems and published research in robotics.

Skills

Programming: Python, C++, MySQL, MongoDB, SQLite3, Basic Assembly Language

Libraries & Frameworks: TensorFlow, PyTorch, OpenCV, Keras, LangChain, LangGraph, Scikit-learn, Pandas, NumPy, Matplotlib, Filterpy, BeautifulSoup, Selenium

AI/ML Expertise: CNN, LSTM, YOLO, DeepSORT, Kalman Filters, NLP, RAG, LLM Integration, Prompt Engineering

Tools & Platforms: Dialogflow CX, Vertex AI, React, FastAPI, Flask, Docker, Git & GitHub, Power BI, Microsoft Excel, AWS, Hostinger

Soft Skills: Team Leadership, Cross-functional Collaboration, Public Speaking, Problem Solving, Agile & Scrum, Technical Communication, Elocution

Experience

AI Engineer

June 2025 – Present

Techs & Tomes

Pune

- Deployed a multilingual skincare chatbot using LangChain and RAG, reducing average query resolution time by ~40%.
- Led complete backend development for the live Formulynx.ai project — designed APIs, database architecture, and AI integration to support real-time formulation analysis and deployment.
- Built AI video translator and summarizer using faster-whisper and Edge TTS, processing 50+ videos with 95%+ transcription accuracy.
- Created scalable web scrapers for skincare data extraction and cleaning, collecting 10,000+ structured product records.
- Designed Skin Vision, a privacy-preserving AI facial skin analyzer with budget-aware recommendations, integrated via React and Tailwind.
- Handled frontend for admin panel and website redevelopment of SkinBB, including API integration and UI implementation under tight deadlines using React, Tailwind CSS, and AI tools.

AI Intern

Nov 2024 – May 2025

ARDE DRDO

Pune

- Designed a custom CNN model for aerial object detection using GELAN activation and KMeans-based anchor box optimization for high accuracy on small moving targets.
- Developed a hybrid Kalman Filter + LSTM model for real-time drone tracking with 99%+ predictive accuracy.
- Built a PDF-based RAG chatbot integrating LLaMA and DeepSeek to answer technical queries from 500+ defense documents.
- Researched autonomous drone-following-drone navigation using YOLO, DeepSORT, SLAM, and D* path planning in dynamic environments.

Smart Traffic Management Intern

Feb 2025 – Mar 2025

Infosys Springboard

Remote

- Developed ANPR and ATCC systems using YOLO, OCR, Python, Flask, and React with helmet detection, heatmaps, crosswalk and signal violation detection.
- Collaborated in an Agile environment with GitHub for version control, delivering all milestones within a 6-week sprint cycle.

Development Intern

June 2024 – Nov 2024

Standard Learner

Remote

- Updated and maintained a mock test platform serving active student users.
- Implemented new features, resolved production issues, and collaborated on chatbot development and AI model integration.

Data Science Intern

May 2024 – Jul 2024

CodeClause

Remote

- Worked on data science pipelines including EDA, feature engineering, and predictive model building using Python and Scikit-learn.

Web Development Intern

Feb 2024 – Mar 2024

TechnoHacks Edutech Ltd.

Remote

- Built and maintained web application features using standard frontend and backend development practices.

Business Analyst Training

Jan 2024 – Feb 2024

Finlatics

Remote

- Completed structured business analyst training covering data interpretation, reporting, and analytical decision-making frameworks.

Projects

RahiAI – Intelligent Tourism Chatbot

- Architected a full-stack tourism chatbot providing personalized travel recommendations and itinerary planning by integrating real-time data APIs.
- **Tech Stack:** React.js · Python · FastAPI · LangGraph · MongoDB

SafeSurf AI – Malicious Website Detector

- Engineered a browser extension providing real-time protection against malicious websites using a Scikit-learn classifier integrated with VirusTotal API, achieving 94%+ classification accuracy.
- **Tech Stack:** Django · Python · Scikit-learn · VirusTotal API · Selenium · Playwright

Human Emulated Robotic Hand

- Developed an IoT flex sensor glove that transmits flex data to replicate hand movements on a robotic replica; published in IJARSCT and presented at ICMETET.
- **Tech Stack:** IoT · Flex Sensors · Arduino · Excel

Multiformat Deepfake Detection

- Built a multimodal deepfake classifier for image, voice, and text with 91%+ accuracy using React frontend and Flask backend.
- **Tech Stack:** React · Flask · Python · Deep Learning

Heart Disease Risk Assessment

- Constructed a GUI-based cardiovascular risk assessment tool using SQLite3 and PySimpleGUI, evaluating risk based on 10+ clinical input parameters.
- **Tech Stack:** Python · SQLite3 · PySimpleGUI

AI/NLP Mini Projects

- Built NLP prediction models, sentiment analysis with K-means clustering, and text classification using NLTK/SpaCy and Scikit-learn; presented Deep Reinforcement Learning concepts and applications.

Clap Switch – IoT Home Automation

- Designed sound-sensing electronic circuits for home automation using IoT and embedded electronics principles.

Research Work

Human Emulated Robotic Hand

Published – IJARSCT · Presented – ICMETET

- Research on affordable neuro-prosthetics using a flex-sensor glove and micro-actuated robotic replica to emulate human hand movements.

Multi-Criteria Decision Making in Mechanical Systems

Ongoing

- Applying Fuzzy-DEMATEL, VIKOR, and TOPSIS for performance evaluation and raw material supplier optimization in collaboration with Mechanical Engineering faculty.

Early Prediction of PCOD/PCOS using Multimodal AI

Ongoing

- Developing a machine learning and computer vision approach combining clinical parameters, ultrasound features, and patient history for early risk detection using multimodal fusion and bias-aware medical ML pipelines.

Theory of Mind in Multi-Agent Reinforcement Learning

Ongoing

- Designing ToM-based cognitive agents capable of modeling opponent beliefs and intentions in MARL environments, including ToM-net, opponent modeling, and cooperative behavior emergence.

Certifications

- **Google AI Essentials** (Google, 2024)
- **Introduction to Generative AI** (Google, 2024)
- **Badge & Webhook** (Google Cloud Skills Boost, 2024)
- **SQL and Relational Databases** (IBM, 2024)
- **Foundations of Cybersecurity** (Google, 2024)
- **Infosys Springboard:** Mastering IoT with Arduino · Introduction to AI · Introduction to Deep Learning · 45+ additional certificates

Achievements

- Winner – 1st place, DATAthon at ISBM COE
- Winner – 1st place, State-level Hackathon at Ajeenkya DY Patil School of Technology, Pune
- 1st Position – Project Competition, ISB&M Bangalore
- 1st Position – Idea Hackathon, ISBM Kolkata
- Top 5% with Elite + Silver Medal – NPTEL Mobile Virtual Reality and Artificial Intelligence Certification
- Second Runner-up – EDathon at Ajinkya DY Patil School of Technology, Pune
- Top 8 – Inceptia National-Level Hackathon at PCCOER, Pune
- Debate winner at multiple intercollegiate events

Extracurricular Activities

- Coordinator, IoT Tech-lab and Research Centre
- Member, Technical Placement Cell ISBM COE; Member, Coding Club; Student Association Anchor and Treasurer
- Mentee, Infosys Springboard Pragati Program
- Active participant in anchoring, elocution, and debating events

Education

B.E. Artificial Intelligence and Data Science
ISBM College of Engineering, Savitribai Phule Pune University, Pune

2022 – 2026
SGPA: 8.62 / 10

HSC
Loyola Junior College, Pune

2022
Percentage: 73.33%

SSC
Modern High School, NCL Campus, Pune

2020
Percentage: 93.40%

Languages

English | Hindi | Marathi